Proof-relevant interpolation for the sequent calculus of bicartesian categories

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Craig interpolation

Craig interpolation:

$$A \to C \quad \Longrightarrow \quad \exists B \ (A \to B) \text{ and } (B \to C) \quad \text{with} \quad \mathrm{var}(B) \subseteq \mathrm{var}(A) \cap \mathrm{var}(C).$$

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Craig interpolation is a fundamental property of logical systems. It has been studied for classical, intermediate, and substructural logics. It is also related to Beth-definability and has applications to model checking and program verification.

Deductive interpolation

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The seqeunt calculus of bicartesian categories

Fix a set At of atoms. Formulae are generated by

$$A, B ::= p \mid \top \mid \bot \mid A \wedge B \mid A \vee B \qquad (p \in \mathrm{At}).$$

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Fix a set At of atoms. Formulae are generated by

$$A, B ::= p \mid \top \mid \bot \mid A \wedge B \mid A \vee B \qquad (p \in \mathrm{At}).$$

The cut-free derivations are generated by the following rules:

$$\frac{}{A \vdash A}\ \mathrm{ax} \qquad \frac{}{A \vdash \top}\ \top\mathrm{R} \qquad \frac{}{\bot \vdash C}\ \bot\mathrm{L}$$

$$\frac{A \vdash C}{A \wedge B \vdash C}\ \wedge\mathrm{L}_1 \qquad \frac{B \vdash C}{A \wedge B \vdash C}\ \wedge\mathrm{L}_2 \qquad \frac{A \vdash C \quad A \vdash D}{A \vdash C \wedge D}\ \wedge\mathrm{R}$$

$$\frac{A \vdash C \quad B \vdash C}{A \vee B \vdash C}\ \vee\mathrm{L} \qquad \frac{A \vdash C}{A \vdash C \vee D}\ \vee\mathrm{R}_1 \qquad \frac{A \vdash D}{A \vdash C \vee D}\ \vee\mathrm{R}_2$$

Cut admissibility

The cut rule has the form

$$\frac{A \overset{f}{\vdash} B \qquad B \overset{f'}{\vdash} C}{A \vdash C}\ \text{cut}$$

For cut-free $f : A \vdash B$ and $f' : B \vdash C$, define the cut-free derivation $\text{cut}(f, f') : A \vdash C$ by lexicographic recursion, inspecting the last rule of f' before the last rule of f.

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$$\dfrac{\overset{f}{A \vdash B} \quad \dfrac{}{B \vdash B}\ \text{ax}}{A \vdash B}\ \text{cut} \quad = \quad \overset{f}{A \vdash B}$$

$$\dfrac{\overset{f}{A \vdash B} \quad \dfrac{}{B \vdash \top}\ \top\text{R}}{A \vdash \top}\ \text{cut} \quad = \quad \dfrac{}{A \vdash \top}\ \top\text{R}$$

Cut admissibility

If the second derivation ends in $\wedge$R, apply cut recursively to its two premises.

$$\dfrac{\overset{f}{A \vdash B} \quad \dfrac{\overset{f_1'}{B \vdash C_1} \quad \overset{f_2'}{B \vdash C_2}}{B \vdash C_1 \wedge C_2}\,\wedge\mathrm{R}}{A \vdash C_1 \wedge C_2}\,\mathrm{cut} \;=\; \dfrac{\dfrac{\overset{f}{A \vdash B} \quad \overset{f_1'}{B \vdash C_1}}{A \vdash C_1}\,\mathrm{cut} \quad \dfrac{\overset{f}{A \vdash B} \quad \overset{f_2'}{B \vdash C_2}}{A \vdash C_2}\,\mathrm{cut}}{A \vdash C_1 \wedge C_2}\,\wedge\mathrm{R}$$

Cut admissibility

If the second derivation ends in $\wedge \mathrm{L}_1$ and the first ends in $\wedge \mathrm{R}$, cut f_1 with f'.

$$\dfrac{\dfrac{\overset{f_1}{A \vdash B_1} \quad \overset{f_2}{A \vdash B_2}}{A \vdash B_1 \wedge B_2}\,\wedge \mathrm{R} \quad \dfrac{\overset{f'}{B_1 \vdash C}}{B_1 \wedge B_2 \vdash C}\,\wedge \mathrm{L}_1}{A \vdash C}\,\mathrm{cut} \quad = \quad \dfrac{\overset{f_1}{A \vdash B_1} \quad \overset{f'}{B_1 \vdash C}}{A \vdash C}\,\mathrm{cut}$$

Properties of the admissible cut

For cut-free derivations $f : A \vdash B$, $f' : B \vdash C$, and $f'' : C \vdash D$,

$$\mathsf{cut}(\mathsf{ax}, f) = f, \qquad \mathsf{cut}(f, \mathsf{ax}) = f,$$

$$\mathsf{cut}(\mathsf{cut}(f, f'), f'') = \mathsf{cut}(f, \mathsf{cut}(f', f'')).$$

The identities and associativity are equalities of cut-free derivations.

Maehara's method

For a cut-free derivation $f : A \vdash C$, we define a function mip that produces an interpolant formula D and two derivations $g : A \vdash D$ and $h : D \vdash C$ such that $\mathrm{var}(D) \subseteq \mathrm{var}(A) \cap \mathrm{var}(C)$.

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For any f, we write $mip(f) = (D, g, h)$, which we call an interpolation triple.

Proof proceeds by induction on derivations.

Maehara's method

Base cases:

f	D	g	h
$\dfrac{}{A \vdash A}\ \mathrm{ax}$	A	$\dfrac{}{A \vdash A}\ \mathrm{ax}$	$\dfrac{}{A \vdash A}\ \mathrm{ax}$
$\dfrac{}{A \vdash \top}\ \top\mathrm{R}$	$\top$	$\dfrac{}{A \vdash \top}\ \top\mathrm{R}$	$\dfrac{}{\top \vdash \top}\ \mathrm{ax}$
$\dfrac{}{\bot \vdash C}\ \bot\mathrm{L}$	$\bot$	$\dfrac{}{\bot \vdash \bot}\ \mathrm{ax}$	$\dfrac{}{\bot \vdash C}\ \bot\mathrm{L}$

Maehara's method

$\underline{\wedge \mathsf{L}_1}$:

$$\frac{\overset{f}{A \vdash C}}{A \wedge B \vdash C}\ \wedge \mathsf{L}_1$$

By I.H., we have $\mathrm{mip}(f) = (D, g : A \vdash D, h : D \vdash C)$, then

$$\mathrm{mip}(\wedge \mathsf{L}_1 f) = (D, \wedge \mathsf{L}_1 g, h).$$

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$$\mathrm{mip}(\wedge \mathsf{L}_1 f) = (D, \wedge \mathsf{L}_1 g, h).$$

$$\frac{\overset{g}{A \vdash D}}{A \wedge B \vdash D}\ \wedge \mathsf{L}_1 \qquad D \overset{h}{\vdash} C$$

Maehara's method

$\underline{\wedge}$R:

$$\dfrac{\overset{f_1}{A \vdash C_1} \quad \overset{f_2}{A \vdash C_2}}{A \vdash C_1 \wedge C_2}\ \wedge\mathrm{R}$$

By I.H., we have:

$$\begin{array}{lll} \mathrm{mip}(f_1) = (D_1, g_1, h_1), & g_1 : A \vdash D_1, & h_1 : D_1 \vdash C_1 \\ \mathrm{mip}(f_2) = (D_2, g_2, h_2), & g_2 : A \vdash D_2, & h_2 : D_2 \vdash C_2. \end{array}$$

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The desired interpolant formula is $D_1 \wedge D_2$:

$$\dfrac{\overset{g_1}{A \vdash D_1} \quad \overset{g_2}{A \vdash D_2}}{A \vdash D_1 \wedge D_2}\ \wedge\mathrm{R} \qquad \dfrac{\dfrac{\overset{h_1}{D_1 \vdash C_1}}{D_1 \wedge D_2 \vdash C_1}\ \wedge\mathrm{L}_1 \quad \dfrac{\overset{h_2}{D_2 \vdash C_2}}{D_1 \wedge D_2 \vdash C_2}\ \wedge\mathrm{L}_2}{D_1 \wedge D_2 \vdash C_1 \wedge C_2}\ \wedge\mathrm{R}$$

Maehara's method

$\underline{\vee \mathrm{L}}$:

$$\dfrac{\overset{f_1}{A_1 \vdash C} \qquad \overset{f_2}{A_2 \vdash C}}{A_1 \vee A_2 \vdash C}\ \vee \mathrm{L}$$

By I.H., we have:

$$\begin{array}{lll} \mathrm{mip}(f_1) = (D_1, g_1, h_1), & g_1 : A_1 \vdash D_1, & h_1 : D_1 \vdash C \\ \mathrm{mip}(f_2) = (D_2, g_2, h_2), & g_2 : A_2 \vdash D_2, & h_2 : D_2 \vdash C. \end{array}$$

Maehara's method

$\vee$L:

$$\dfrac{\overset{f_1}{A_1 \vdash C} \quad \overset{f_2}{A_2 \vdash C}}{A_1 \vee A_2 \vdash C}\ \vee\mathrm{L}$$

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The desired interpolant formula is $D_1 \vee D_2$:

$$\dfrac{\dfrac{\overset{g_1}{A_1 \vdash D_1}}{A_1 \vdash D_1 \vee D_2}\ \vee\mathrm{R}_1 \quad \dfrac{\overset{g_2}{A_2 \vdash D_2}}{A_2 \vdash D_1 \vee D_2}\ \vee\mathrm{R}_2}{A_1 \vee A_2 \vdash D_1 \vee D_2}\ \vee\mathrm{L} \qquad \dfrac{\overset{h_1}{D_1 \vdash C} \quad \overset{h_2}{D_2 \vdash C}}{D_1 \vee D_2 \vdash C}\ \vee\mathrm{L}$$

Proof equivalence

Let $\mathrel{\mathring{=}}$ be the smallest congruence relation containing the following equations. The η-expansions are

$$\frac{}{A \wedge B \vdash A \wedge B}\,\mathsf{ax} \quad \mathrel{\mathring{=}} \quad \dfrac{\dfrac{\dfrac{}{A \vdash A}\,\mathsf{ax}}{A \wedge B \vdash A}\,\wedge\mathsf{L}_1 \quad \dfrac{\dfrac{}{B \vdash B}\,\mathsf{ax}}{A \wedge B \vdash B}\,\wedge\mathsf{L}_2}{A \wedge B \vdash A \wedge B}\,\wedge\mathsf{R}$$

$$\frac{}{A \vee B \vdash A \vee B}\,\mathsf{ax} \quad \mathrel{\mathring{=}} \quad \dfrac{\dfrac{\dfrac{}{A \vdash A}\,\mathsf{ax}}{A \vdash A \vee B}\,\vee\mathsf{R}_1 \quad \dfrac{\dfrac{}{B \vdash B}\,\mathsf{ax}}{B \vdash A \vee B}\,\vee\mathsf{R}_2}{A \vee B \vdash A \vee B}\,\vee\mathsf{L}$$

Proof equivalence

The unit equations are

$$\overset{f}{A \vdash \top} \quad \doteq \quad \frac{}{A \vdash \top}\,\top\mathrm{R}$$

$$\overset{f}{\bot \vdash A} \quad \doteq \quad \frac{}{\bot \vdash A}\,\bot\mathrm{L}$$

The commuting equations of $\wedge\mathrm{R}$ and $\wedge\mathrm{L}_i$ ($i \in \{1, 2\}$):

$$\frac{\dfrac{\overset{f}{A_i \vdash C_1} \quad \overset{g}{A_i \vdash C_2}}{A_i \vdash C_1 \wedge C_2}\,\wedge\mathrm{R}}{A_1 \wedge A_2 \vdash C_1 \wedge C_2}\,\wedge\mathrm{L}_i \quad \doteq \quad \frac{\dfrac{\overset{f}{A_i \vdash C_1}}{A_1 \wedge A_2 \vdash C_1}\,\wedge\mathrm{L}_i \quad \dfrac{\overset{g}{A_i \vdash C_2}}{A_1 \wedge A_2 \vdash C_2}\,\wedge\mathrm{L}_i}{A_1 \wedge A_2 \vdash C_1 \wedge C_2}\,\wedge\mathrm{R}$$

Proof equivalence

The commuting equations ($i, j \in \{1, 2\}$):

$$\dfrac{\dfrac{\overset{f}{A_i \vdash C_j}}{A_i \vdash C_1 \vee C_2}\,\vee \mathrm{R}_j}{A_1 \wedge A_2 \vdash C_1 \vee C_2}\,\wedge \mathrm{L}_i \quad \stackrel{\circ}{=} \quad \dfrac{\dfrac{\overset{f}{A_i \vdash C_j}}{A_1 \wedge A_2 \vdash C_j}\,\wedge \mathrm{L}_i}{A_1 \wedge A_2 \vdash C_1 \vee C_2}\,\vee \mathrm{R}_j$$

$$\dfrac{\dfrac{\overset{f}{A_1 \vdash C_j}}{A_1 \vdash C_1 \vee C_2}\,\vee \mathrm{R}_j \quad \dfrac{\overset{g}{A_2 \vdash C_j}}{A_2 \vdash C_1 \vee C_2}\,\vee \mathrm{R}_j}{A_1 \vee A_2 \vdash C_1 \vee C_2}\,\vee \mathrm{L} \quad \stackrel{\circ}{=} \quad \dfrac{\dfrac{\overset{f}{A_1 \vdash C_j} \quad \overset{g}{A_2 \vdash C_j}}{A_1 \vee A_2 \vdash C_j}\,\vee \mathrm{L}}{A_1 \vee A_2 \vdash C_1 \vee C_2}\,\vee \mathrm{R}_j$$

Proof equivalence

The commuting equations:

$$\dfrac{\dfrac{\overset{f_1}{A_1 \vdash C_1} \quad \overset{g_1}{A_1 \vdash C_2}}{A_1 \vdash C_1 \wedge C_2}\wedge\mathrm{R} \quad \dfrac{\overset{f_2}{A_2 \vdash C_1} \quad \overset{g_2}{A_2 \vdash C_2}}{A_2 \vdash C_1 \wedge C_2}\wedge\mathrm{R}}{A_1 \vee A_2 \vdash C_1 \wedge C_2}\vee\mathrm{L}$$

$$\stackrel{\circ}{=} \quad \dfrac{\dfrac{\overset{f_1}{A_1 \vdash C_1} \quad \overset{f_2}{A_2 \vdash C_1}}{A_1 \vee A_2 \vdash C_1}\vee\mathrm{L} \quad \dfrac{\overset{g_1}{A_1 \vdash C_2} \quad \overset{g_2}{A_2 \vdash C_2}}{A_1 \vee A_2 \vdash C_2}\vee\mathrm{L}}{A_1 \vee A_2 \vdash C_1 \wedge C_2}\wedge\mathrm{R}$$

Interpolation triples

An interpolation triple is

$$(D, g, h), \qquad g : A \vdash D, \qquad h : D \vdash C.$$

satisfying the variable condition

$$\mathrm{var}(D) \subseteq \mathrm{var}(A) \cap \mathrm{var}(C).$$

It factors a cut-free derivation $f : A \vdash C$ when $\mathrm{cut}(g, h) \stackrel{\circ}{=} f$.

Proof-relevant interpolation

Traditionally, Maehara interpolation only states the *mere existence* of an interpolation triple (D, g, h).

Proof-relevant interpolation studies its relationship to admissible cut and equivalence of proofs. Precisely, it states that cut is a left inverse of the interpolation procedure. This line of work was initiated in the 90s by Čubrić and recently revived by Saurin.

Cut as a left inverse of mip

$$\mathrm{mip}(f) = (D, g, h) \qquad \Longrightarrow \qquad \mathrm{cut}(g, h) \mathrel{\mathring{=}} f.$$

First apply mip to a derivation f, then cut the two resulting derivations. This gives back f up to $\mathring{=}$: cut is a left inverse of mip.

Proof proceeds by induction on f. We show the $\wedge$R and $\vee$L cases.

Cut as a left inverse of mip

$\underline{\wedge \mathrm{R}}$: if $f = \wedge \mathrm{R}(f_1, f_2)$, then we have $\mathrm{mip}(f_k) = (D_k, g_k, h_k)$ for $k = 1, 2$.

We defined $\mathrm{mip}(f) = (D_1 \wedge D_2, g, h)$, where

$$g = \wedge \mathrm{R}(g_1, g_2), \qquad h = \wedge \mathrm{R}(\wedge \mathrm{L}_1 h_1, \wedge \mathrm{L}_2 h_2).$$

We calculate:

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$$g = \wedge\mathrm{R}(g_1, g_2), \qquad h = \wedge\mathrm{R}(\wedge\mathrm{L}_1 h_1, \wedge\mathrm{L}_2 h_2).$$

We calculate:

$$\begin{aligned}
&\mathrm{cut}(g, h) \\
&= \mathrm{cut}(\wedge\mathrm{R}(g_1, g_2), \wedge\mathrm{R}(\wedge\mathrm{L}_1 h_1, \wedge\mathrm{L}_2 h_2)) && \text{(unfold definitions)} \\
&= \wedge\mathrm{R}\big(\mathrm{cut}(\wedge\mathrm{R}(g_1, g_2), \wedge\mathrm{L}_1 h_1), \\
&\qquad\qquad \mathrm{cut}(\wedge\mathrm{R}(g_1, g_2), \wedge\mathrm{L}_2 h_2)\big) && \text{(definition of cut)} \\
&= \wedge\mathrm{R}(\mathrm{cut}(g_1, h_1), \mathrm{cut}(g_2, h_2)) && \text{(definition of cut)} \\
&\stackrel{\circ}{=} \wedge\mathrm{R}(f_1, f_2) = f && \text{(I.H. on } f_1 \text{ and } f_2\text{)}.
\end{aligned}$$

Cut as a left inverse of mip

$\underline{\vee \mathsf{L}}$: if $f = \vee\mathsf{L}(f_1, f_2)$, then we have $\mathrm{mip}(f_k) = (D_k, g_k, h_k)$ for $k = 1, 2$.

We defined $\mathrm{mip}(f) = (D_1 \vee D_2, g, h)$, where

$$g = \vee\mathsf{L}(\vee\mathsf{R}_1 g_1, \vee\mathsf{R}_2 g_2), \qquad h = \vee\mathsf{L}(h_1, h_2).$$

We calculate:

Cut as a left inverse of mip

$\underline{\vee\mathsf{L}}$: if $f = \vee\mathsf{L}(f_1, f_2)$, then we have $\mathrm{mip}(f_k) = (D_k, g_k, h_k)$ for $k = 1, 2$.

We defined $\mathrm{mip}(f) = (D_1 \vee D_2, g, h)$, where

$$g = \vee\mathsf{L}(\vee\mathsf{R}_1 g_1, \vee\mathsf{R}_2 g_2), \qquad h = \vee\mathsf{L}(h_1, h_2).$$

We calculate:

$$\begin{aligned}
&\mathrm{cut}(g, h) \\
&= \mathrm{cut}(\vee\mathsf{L}(\vee\mathsf{R}_1 g_1, \vee\mathsf{R}_2 g_2), \vee\mathsf{L}(h_1, h_2)) && \text{(unfold definitions)} \\
&= \vee\mathsf{L}\big(\mathrm{cut}(\vee\mathsf{R}_1 g_1, \vee\mathsf{L}(h_1, h_2)), \\
&\qquad\qquad \mathrm{cut}(\vee\mathsf{R}_2 g_2, \vee\mathsf{L}(h_1, h_2))\big) && \text{(definition of cut)} \\
&= \vee\mathsf{L}(\mathrm{cut}(g_1, h_1), \mathrm{cut}(g_2, h_2)) && \text{(definition of cut)} \\
&\stackrel{\circ}{=} \vee\mathsf{L}(f_1, f_2) = f && \text{(I.H. on } f_1 \text{ and } f_2\text{).}
\end{aligned}$$

Well-definedness

We want mip to interact well with proof equivalence: equivalent derivations should be sent to equivalent interpolation triples.

$$f \stackrel{\circ}{=} f' \qquad \Longrightarrow \qquad \mathsf{mip}(f) \sim \mathsf{mip}(f').$$

Well-definedness

We want mip to interact well with proof equivalence: equivalent derivations should be sent to equivalent interpolation triples.

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To define equivalence of interpolation triples, consider how mip acts on the commuting equation between $\vee$L and $\wedge$R.

Well-definedness

The two derivations are equivalent:

$$\dfrac{\dfrac{\dfrac{\overset{f_1}{A_1 \vdash C_1} \quad \overset{g_1}{A_1 \vdash C_2}}{A_1 \vdash C_1 \wedge C_2}\,\wedge\mathrm{R} \quad \dfrac{\overset{f_2}{A_2 \vdash C_1} \quad \overset{g_2}{A_2 \vdash C_2}}{A_2 \vdash C_1 \wedge C_2}\,\wedge\mathrm{R}}{A_1 \vee A_2 \vdash C_1 \wedge C_2}\,\vee\mathrm{L}$$

$$\mathrel{\mathring{=}} \dfrac{\dfrac{\overset{f_1}{A_1 \vdash C_1} \quad \overset{f_2}{A_2 \vdash C_1}}{A_1 \vee A_2 \vdash C_1}\,\vee\mathrm{L} \quad \dfrac{\overset{g_1}{A_1 \vdash C_2} \quad \overset{g_2}{A_2 \vdash C_2}}{A_1 \vee A_2 \vdash C_2}\,\vee\mathrm{L}}{A_1 \vee A_2 \vdash C_1 \wedge C_2}\,\wedge\mathrm{R}$$

Well-definedness

Interpolating the four premise derivations gives the following four triples:

$$\mathsf{mip}(f_1) = (P_1, g_1' : A_1 \vdash P_1, h_1' : P_1 \vdash C_1),$$

$$\mathsf{mip}(g_1) = (Q_1, g_1'' : A_1 \vdash Q_1, h_1'' : Q_1 \vdash C_2),$$

$$\mathsf{mip}(f_2) = (P_2, g_2' : A_2 \vdash P_2, h_2' : P_2 \vdash C_1),$$

$$\mathsf{mip}(g_2) = (Q_2, g_2'' : A_2 \vdash Q_2, h_2'' : Q_2 \vdash C_2).$$

Well-definedness

For the upper derivation, $\wedge$R first gives $P_i \wedge Q_i$ in each premise; $\vee$L then gives the triple (D', g', h'):

$$D' = (P_1 \wedge Q_1) \vee (P_2 \wedge Q_2),$$

$$g' = \vee\mathrm{L}(\vee\mathrm{R}_1 \wedge\mathrm{R}(g_1', g_1''), \vee\mathrm{R}_2 \wedge\mathrm{R}(g_2', g_2'')),$$

$$h' = \vee\mathrm{L}(\wedge\mathrm{R}(\wedge\mathrm{L}_1 h_1', \wedge\mathrm{L}_2 h_1''), \wedge\mathrm{R}(\wedge\mathrm{L}_1 h_2', \wedge\mathrm{L}_2 h_2'')).$$

Well-definedness

For the upper derivation, $\wedge$R first gives $P_i \wedge Q_i$ in each premise; $\vee$L then gives the triple (D', g', h'):

$$D' = (P_1 \wedge Q_1) \vee (P_2 \wedge Q_2),$$
$$g' = \vee\mathrm{L}(\vee\mathrm{R}_1 \wedge\mathrm{R}(g_1', g_1''), \vee\mathrm{R}_2 \wedge\mathrm{R}(g_2', g_2'')),$$
$$h' = \vee\mathrm{L}(\wedge\mathrm{R}(\wedge\mathrm{L}_1 h_1', \wedge\mathrm{L}_2 h_1''), \wedge\mathrm{R}(\wedge\mathrm{L}_1 h_2', \wedge\mathrm{L}_2 h_2'')).$$

For the lower derivation, $\vee$L first gives $P_1 \vee P_2$ and $Q_1 \vee Q_2$; $\wedge$R then gives the triple (D'', g'', h''):

$$D'' = (P_1 \vee P_2) \wedge (Q_1 \vee Q_2),$$
$$g'' = \wedge\mathrm{R}(\vee\mathrm{L}(\vee\mathrm{R}_1 g_1', \vee\mathrm{R}_2 g_2'), \vee\mathrm{L}(\vee\mathrm{R}_1 g_1'', \vee\mathrm{R}_2 g_2'')),$$
$$h'' = \wedge\mathrm{R}(\wedge\mathrm{L}_1 \vee\mathrm{L}(h_1', h_2'), \wedge\mathrm{L}_2 \vee\mathrm{L}(h_1'', h_2'')).$$

Well-definedness

Only the direction from $(P_1 \wedge Q_1) \vee (P_2 \wedge Q_2)$ to $(P_1 \vee P_2) \wedge (Q_1 \vee Q_2)$ is derivable:

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{P_1 \vdash P_1}\,\text{ax}}{P_1 \wedge Q_1 \vdash P_1}\,\wedge\text{L}_1}{P_1 \wedge Q_1 \vdash P_1 \vee P_2}\,\vee\text{R}_1 \quad \dfrac{\dfrac{\dfrac{}{Q_1 \vdash Q_1}\,\text{ax}}{P_1 \wedge Q_1 \vdash Q_1}\,\wedge\text{L}_2}{P_1 \wedge Q_1 \vdash Q_1 \vee Q_2}\,\vee\text{R}_1}{P_1 \wedge Q_1 \vdash (P_1 \vee P_2) \wedge (Q_1 \vee Q_2)}\,\wedge\text{R} \quad \dfrac{\dfrac{\dfrac{\dfrac{}{P_2 \vdash P_2}\,\text{ax}}{P_2 \wedge Q_2 \vdash P_2}\,\wedge\text{L}_1}{P_2 \wedge Q_2 \vdash P_1 \vee P_2}\,\vee\text{R}_2 \quad \dfrac{\dfrac{\dfrac{}{Q_2 \vdash Q_2}\,\text{ax}}{P_2 \wedge Q_2 \vdash Q_2}\,\wedge\text{L}_2}{P_2 \wedge Q_2 \vdash Q_1 \vee Q_2}\,\vee\text{R}_2}{P_2 \wedge Q_2 \vdash (P_1 \vee P_2) \wedge (Q_1 \vee Q_2)}\,\wedge\text{R}}{(P_1 \wedge Q_1) \vee (P_2 \wedge Q_2) \vdash (P_1 \vee P_2) \wedge (Q_1 \vee Q_2)}\,\vee\text{L}$$

The reverse sequent has no derivation in general:

$$(P_1 \vee P_2) \wedge (Q_1 \vee Q_2) \vdash (P_1 \wedge Q_1) \vee (P_2 \wedge Q_2).$$

Equivalence of interpolation triples

Definition

Two interpolation triples are equivalent, written

$$(D, g, h) \sim (D', g', h'),$$

if they belong to the symmetric transitive closure of the following relation: there exists a derivation $t : D \vdash D'$ such that

$$\mathrm{cut}(g, t) \stackrel{\circ}{=} g', \qquad h \stackrel{\circ}{=} \mathrm{cut}(t, h').$$

We write the morphism defined by t as

$$(D, g, h) \xrightarrow{t} (D', g', h').$$

Equivalence of interpolation triples

The relation $\sim$ is given by finite zig-zags of morphisms of interpolation triples, for example:

$$(D, g, h) \xrightarrow{t_1} (D_1, g_1, h_1) \xleftarrow{t_2} (D_2, g_2, h_2) \xrightarrow{t_3} \cdots \xleftarrow{t_n} (D', g', h').$$

Every arrow satisfies the two equations defining a morphism in its direction.

The identity $\mathrm{ax} : D \vdash D$ gives $(D, g, h) \xrightarrow{\mathrm{ax}} (D, g, h)$, so the relation is reflexive.

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Every triple in the zig-zag is an interpolation triple, so its middle formula satisfies the variable condition.

Well-definedness

The derivation $t : (P_1 \wedge Q_1) \vee (P_2 \wedge Q_2) \vdash (P_1 \vee P_2) \wedge (Q_1 \vee Q_2)$ displayed above is

$$t = \vee\mathsf{L}(\wedge\mathsf{R}(\vee\mathsf{R}_1 \wedge \mathsf{L}_1\mathsf{ax}, \vee\mathsf{R}_1 \wedge \mathsf{L}_2\mathsf{ax}),$$
$$\wedge\mathsf{R}(\vee\mathsf{R}_2 \wedge \mathsf{L}_1\mathsf{ax}, \vee\mathsf{R}_2 \wedge \mathsf{L}_2\mathsf{ax})).$$

For the interpolation triples (D', g', h') and (D'', g'', h''), we calculate

$$\mathsf{cut}(g', t) \mathrel{\mathring{=}} g'', \qquad h' \mathrel{\mathring{=}} \mathsf{cut}(t, h'').$$

Well-definedness

To show $\mathsf{cut}(g', t) \mathrel{\mathring{=}} g''$, we calculate:

$$
\begin{aligned}
&\mathsf{cut}(g', t) \\
&= \mathsf{cut}\big(\vee\mathsf{L}(\vee\mathsf{R}_1\wedge\mathsf{R}(g_1', g_1''), \vee\mathsf{R}_2\wedge\mathsf{R}(g_2', g_2'')), t\big) && \text{(unfold } g'\text{)} \\
&= \vee\mathsf{L}\big(\wedge\mathsf{R}(\vee\mathsf{R}_1\mathsf{cut}(g_1', \mathsf{ax}), \vee\mathsf{R}_1\mathsf{cut}(g_1'', \mathsf{ax})), \\
&\qquad\qquad \wedge\mathsf{R}(\vee\mathsf{R}_2\mathsf{cut}(g_2', \mathsf{ax}), \vee\mathsf{R}_2\mathsf{cut}(g_2'', \mathsf{ax}))\big) && \text{(unfold } t\text{, definition of cut)} \\
&= \vee\mathsf{L}\big(\wedge\mathsf{R}(\vee\mathsf{R}_1 g_1', \vee\mathsf{R}_1 g_1''), \wedge\mathsf{R}(\vee\mathsf{R}_2 g_2', \vee\mathsf{R}_2 g_2'')\big) && (\mathsf{cut}(f, \mathsf{ax}) = f) \\
&\mathrel{\mathring{=}} \wedge\mathsf{R}\big(\vee\mathsf{L}(\vee\mathsf{R}_1 g_1', \vee\mathsf{R}_2 g_2'), \\
&\qquad\qquad \vee\mathsf{L}(\vee\mathsf{R}_1 g_1'', \vee\mathsf{R}_2 g_2'')\big) && (\vee\mathsf{L} \text{ and } \wedge\mathsf{R} \text{ commute}) \\
&= g''. && \text{(definition of } g''\text{)}
\end{aligned}
$$

Well-definedness

To show $h' \mathrel{\mathring{=}} \mathsf{cut}(t, h'')$, use

$$\mathsf{cut}(\wedge \mathsf{L}_i f, h) \mathrel{\mathring{=}} \wedge \mathsf{L}_i \mathsf{cut}(f, h), \qquad \mathsf{cut}(\mathsf{ax}, h) = h.$$

$\mathsf{cut}(t, h'')$

$= \mathsf{cut}\big(t, \wedge \mathsf{R}(\wedge \mathsf{L}_1 \vee \mathsf{L}(h_1', h_2'), \wedge \mathsf{L}_2 \vee \mathsf{L}(h_1'', h_2''))\big)$ (unfold h'')

$= \wedge \mathsf{R}\big(\mathsf{cut}(t, \wedge \mathsf{L}_1 \vee \mathsf{L}(h_1', h_2')),$ (definition of cut)
$\qquad \mathsf{cut}(t, \wedge \mathsf{L}_2 \vee \mathsf{L}(h_1'', h_2''))\big)$

$\mathrel{\mathring{=}} \wedge \mathsf{R}\big(\vee \mathsf{L}(\wedge \mathsf{L}_1 h_1', \wedge \mathsf{L}_1 h_2'),$ (unfold t, definition of cut, equations above)
$\qquad \vee \mathsf{L}(\wedge \mathsf{L}_2 h_1'', \wedge \mathsf{L}_2 h_2'')\big)$

$\mathrel{\mathring{=}} \vee \mathsf{L}\big(\wedge \mathsf{R}(\wedge \mathsf{L}_1 h_1', \wedge \mathsf{L}_2 h_1''),$ ($\vee$L and $\wedge$R commute)
$\qquad \wedge \mathsf{R}(\wedge \mathsf{L}_1 h_2', \wedge \mathsf{L}_2 h_2'')\big)$

$= h'.$ (definition of h')

Hence $(D', g', h') \sim (D'', g'', h'')$.

Well-definedness

Theorem

For $f, f' : A \vdash C$,

$$f \stackrel{\circ}{=} f' \qquad \Longrightarrow \qquad \mathsf{mip}(f) \sim \mathsf{mip}(f')$$

Proof proceeds by induction on the generating equations of $\stackrel{\circ}{=}$.

mip as a left inverse of cut

We are also interested in mip as an inverse of cut. We have seen that applying mip and then cut recovers the original derivation up to $\overset{\circ}{=}$. What happens when we apply them in the other order?

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Starting with an interpolation triple (D, g, h), where $g : A \vdash D$ and $h : D \vdash C$, we cut g with h and then apply mip. We want to recover an equivalent interpolation triple:

$$(D, g, h) \xrightarrow{\text{cut}} \text{cut}(g, h) \xrightarrow{\text{mip}} \text{mip}(\text{cut}(g, h)).$$

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Theorem
For every interpolation triple (D, g, h)*,*

$$\text{mip}(\text{cut}(g, h)) \sim (D, g, h).$$

Thus mip *is a left inverse of cut, up to* $\sim$*.*

Proof proceeds by lexicographic induction, inspecting h first as in cut.

Interpolation triples factoring the same derivation

Suppose (D, g, h) and (D', g', h') both factor f:

$$\mathrm{cut}(g, h) \doteq f, \qquad \mathrm{cut}(g', h') \doteq f.$$

Applying mip after cut and using well-definedness gives

$$\begin{aligned}(D, g, h) &\sim \mathrm{mip}(\mathrm{cut}(g, h)) \\ &\sim \mathrm{mip}(f) \\ &\sim \mathrm{mip}(\mathrm{cut}(g', h')) \\ &\sim (D', g', h').\end{aligned}$$

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Any two interpolation triples factoring f are therefore $\sim$-equivalent. In other words, interpolation triples factoring a derivation f are unique up to zig-zag.

What we have shown

Maehara's method factors a cut-free derivation $f : A \vdash C$:

$$\mathrm{mip}(f) = (D, g, h), \qquad A \xrightarrow{g} D \xrightarrow{h} C, \qquad \mathrm{var}(D) \subseteq \mathrm{var}(A) \cap \mathrm{var}(C).$$

- Cutting g with h gives back the derivation: $\mathrm{cut}(g, h) \stackrel{\circ}{=} f$.
- Equivalent derivations give equivalent interpolation triples: $f \stackrel{\circ}{=} f' \Longrightarrow \mathrm{mip}(f) \sim \mathrm{mip}(f')$.
- Interpolating after cut gives back an equivalent triple: $\mathrm{mip}(\mathrm{cut}(g, h)) \sim (D, g, h)$.

A categorical interpretation

We have seen how derivations, cut, and mip interact. Categorically, we know that

$$\text{derivations up to } \overset{\circ}{=} \quad \longleftrightarrow \quad \text{morphisms},$$

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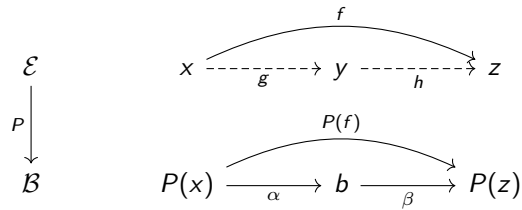
$$\text{cut} \quad \longleftrightarrow \quad \text{composition}.$$

What about mip and the proof-relevant interpolation properties?

Interpolation gives a factorization of a derivation, and interpolation triples factoring the same derivation are equivalent up to a zig-zag. To investigate this categorically, let us first recall Conduché fibrations.

Conduché fibration

Let $P : \mathcal{E} \to \mathcal{B}$ be a functor. Given $f : x \to z$ and $P(f) = \beta \circ \alpha$, a lifting is a factorization $f = h \circ g$ with $P(y) = b$, $P(g) = \alpha$, and $P(h) = \beta$:



The functor P is a Conduché fibration if every such factorization has a lifting, *unique up to zig-zag*: there is a zig-zag of morphisms of liftings between any two liftings.

This suggests a possible connection with proof-relevant interpolation.

The functor to **Rel**

We choose **Rel** as the base category and the free bicartesian category on At as the upper category.

In the upper category, objects are formulas and morphisms are $\overset{\circ}{=}$-classes of cut-free derivations. Identity is represented by ax, and composition by admissible cut.

The functor to **Rel**

Fix a finite set $V(p)$ for every atom p. Use tagged disjoint union, with injections inl and inr. Define

$$[\![p]\!] = V(p), \qquad [\![\top]\!] = [\![\bot]\!] = \varnothing,$$
$$[\![A \wedge B]\!] = [\![A \vee B]\!] = [\![A]\!] + [\![B]\!].$$

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For a cut-free $f : A \vdash C$, define $[\![f]\!] \subseteq [\![A]\!] \times [\![C]\!]$ by the following clauses. Axiom is the diagonal and both unit rules are empty. The four unary rules add the corresponding source or target tag.

$$[\![\wedge \mathsf{L}_1 f]\!] = \{(\mathsf{inl}(a), c) : (a, c) \in [\![f]\!]\}.$$

The functor to **Rel**

The binary clauses are

$$\begin{aligned}[\![\wedge \mathrm{R}(f_1, f_2)]\!] = &\{(a, \mathrm{inl}(c)) : (a, c) \in [\![f_1]\!]\} \\ &\cup \{(a, \mathrm{inr}(e)) : (a, e) \in [\![f_2]\!]\},\end{aligned}$$

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Proof equivalence is sound and

$$[\![\mathrm{cut}(f, f')]\!] = [\![f']\!] \circ [\![f]\!].$$

Formula interpretation and derivation interpretation define a functor to **Rel**.

Maehara interpolation and factorizations

Does $\mathrm{mip}(f)$ lift any given factorization of $[\![f]\!]$?

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Let $V(p) = \{*\}$ and $f = \mathrm{ax}$. Consider the factorization through $B = \{0, 1\}$:

$$\alpha = \{(*, 0), (*, 1)\}, \qquad \beta = \{(0, *), (1, *)\}, \qquad \beta \circ \alpha = [\![f]\!].$$

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Maehara interpolation gives

$$\mathrm{mip}(\mathrm{ax}) = (p, \mathrm{ax}, \mathrm{ax}), \qquad [\![p]\!] = \{*\}.$$

The interpreted middle formula is a singleton, whereas B has two elements. Thus this Maehara triple does not lift the given factorization.

We therefore look for a class of factorizations for which mip satisfies a weaker property.

Cofunctional/functional factorizations

Cofunctional/functional factorizations are natural candidates: if $\mathrm{mip}(f) = (D, g, h)$, then $[\![g]\!]$ is cofunctional and $[\![h]\!]$ is functional.

A relation is functional when each source element is related to exactly one target element. It is cofunctional when each target element is related to exactly one source element.

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Thus a cofunctional/functional factorization

$$[\![A]\!] \xrightarrow{\alpha} B \xrightarrow{\beta} [\![C]\!], \qquad \beta \circ \alpha = [\![f]\!],$$

is equivalently a functional span

$$[\![A]\!] \xleftarrow{\alpha^\circ} B \xrightarrow{\beta} [\![C]\!],$$

where α° is the converse relation.

Maehara interpolation and factorizations

Even cofunctional/functional factorizations need not be lifted by mip.

Let $V(p) = \{*\}$ and $f = \mathrm{ax}$. Take the surjection

$$k : B \to [\![p]\!], \qquad B = \{*, 0\}, \qquad k(*) = k(0) = *.$$

Put $\alpha = k^\circ$ and $\beta = k$. Then α is cofunctional, β is functional, moreover

$$\beta \circ \alpha = k \circ k^\circ = 1_{[\![p]\!]} = [\![f]\!].$$

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However,

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In this example, k is a morphism from (B, α, β) to $([\![p]\!], [\![\mathsf{ax}]\!], [\![\mathsf{ax}]\!])$.

A weak Conduché-like property

Let $f : A \vdash C$ be a derivation, and write $\mathrm{mip}(f) = (D, g, h)$.

Theorem

Every cofunctional/functional factorization

$$[\![A]\!] \xrightarrow{\alpha} B \xrightarrow{\beta} [\![C]\!], \qquad [\![f]\!] = \beta \circ \alpha,$$

admits a unique surjective function $k : B \to [\![D]\!]$ *satisfying*

$$[\![g]\!] = k \circ \alpha, \qquad \beta = [\![h]\!] \circ k.$$

Notice also that $\mathrm{mip}(f)$ is unique up to zig-zag.
Consequently, the interpolation triple obtained from mip is terminal in the category of cofunctional/functional factorizations of $[\![f]\!]$ in **Rel**.

Concluding remarks

- We established that admissible cut is a left inverse of mip up to $\overset{\circ}{=}$, and mip is a left inverse of cut up to $\sim$.
- We demonstrated that mip is well-defined wrt. $\overset{\circ}{=}$, using zig-zags of morphisms to define equivalence of interpolation triples.
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